

MACHINE LEARNING



CLASSICAL MACHINE LEARNING

Data is pre-categorized
or numerical

SUPERVISED

Predict
a category

CLASSIFICATION

Separate the cats from
the dogs



Predict
a number

REGRESSION

Predict the price of a
house



Data is not labeled
in any way

UNSUPERVISED

Divide
by similarity

CLUSTERING

Divide customers into similar
groups



Identify sequences

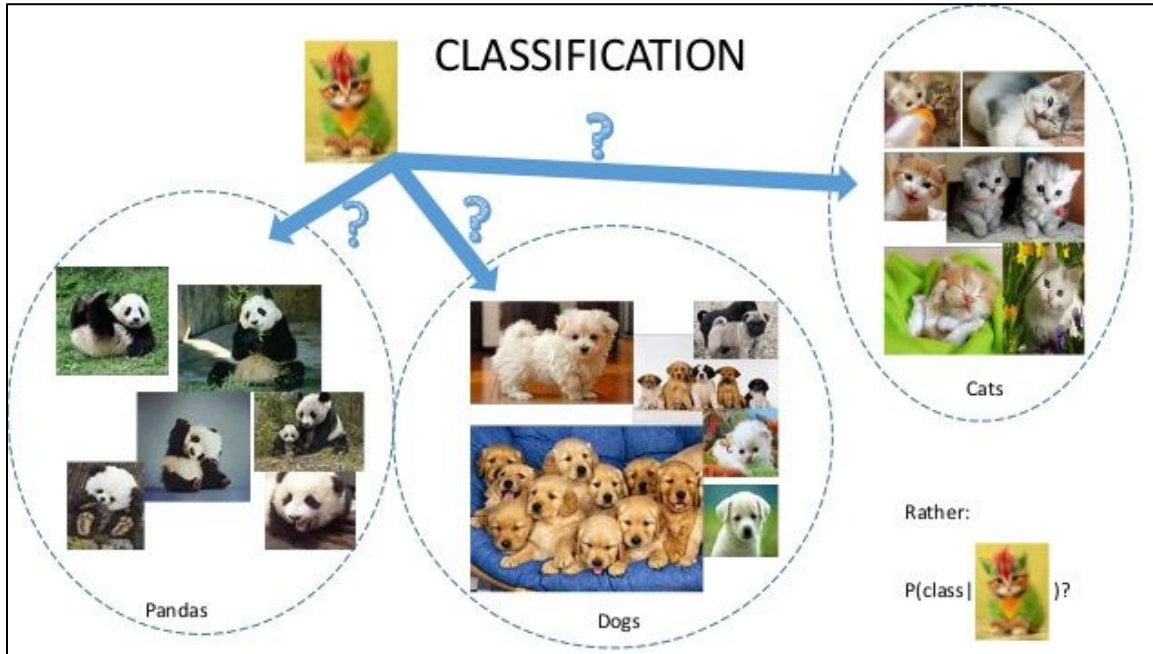
ASSOCIATION

Find items that are often purchased
together



Classification

Assigning objects to one of several predefined categories or class labels



Terminology

Attributes / Features

Objects / Records /

Data Points

Name	Body Temp	Skin Cover	Gives Birth	Aquatic	Aerial	Has Legs	Hibernates	Class Label / Target Value
human	Warm-blooded	hair	yes	no	no	yes	no	mammal
python	Cold-blooded	scales	no	no	no	no	yes	reptile
salmon	Cold-blooded	scales	no	yes	no	no	no	fish
whale	Warm-blooded	hair	yes	yes	no	no	no	mammal
bat	Warm-blooded	hair	yes	no	yes	yes	yes	mammal
salamander	Cold-blooded	none	no	semi	no	yes	yes	amphibian
penguin	Warm-blooded	feathers	no	semi	no	yes	no	bird

Classification (or Prediction)

Descriptive Modeling: explain, describe, summarize the data

Ex: What features define a mammal, reptile, bird, fish, amphibian

Name	Body Temp	Skin Cover	Gives Birth	Aquatic	Aerial	Has Legs	Hibernates	Class Label
human	Warm-blooded	hair	yes	no	no	yes	no	mammal
python	Cold-blooded	scales	no	no	no	no	yes	reptile
salmon	Cold-blooded	scales	no	yes	no	no	no	fish
whale	Warm-blooded	hair	yes	yes	no	no	no	mammal
bat	Warm-blooded	hair	yes	no	yes	yes	yes	mammal
salamander	Cold-blooded	none	no	semi	no	yes	yes	amphibian
penguin	Warm-blooded	feathers	no	semi	no	yes	no	bird

Predictive Modeling: predict the class of a new or unknown record

gila monster	Cold-blooded	scales	no	no	no	yes	yes	?
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Example

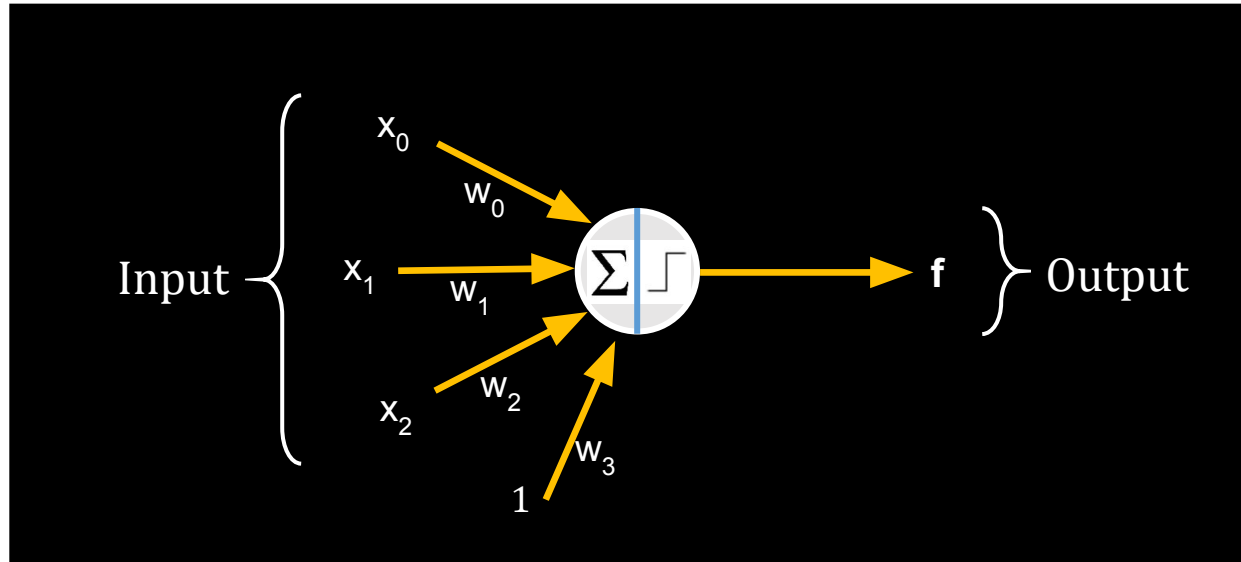
Training data:

Chills	Congestion	Temperature (F)	Flu
Y	N	102	N
Y	Y	98	Y
Y	Y	101	Y
N	Y	99	Y
N	N	97	N
N	Y	98	N
Y	Y	100	Y

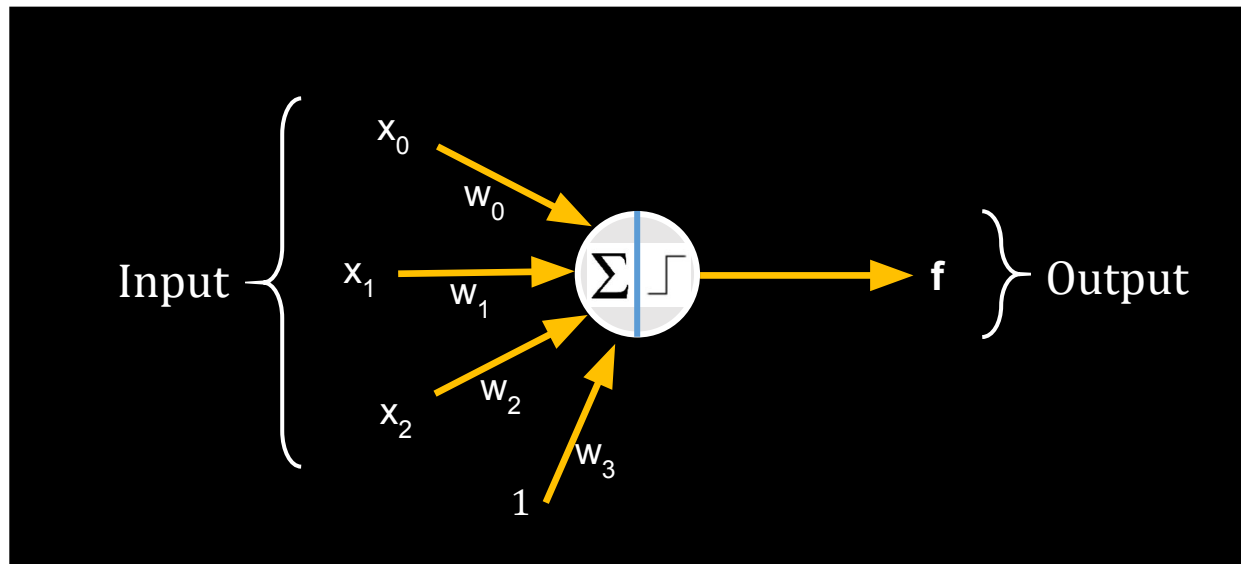
Do you think a patient with the following symptoms has the flu?

Y	N	100	?
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Perceptron

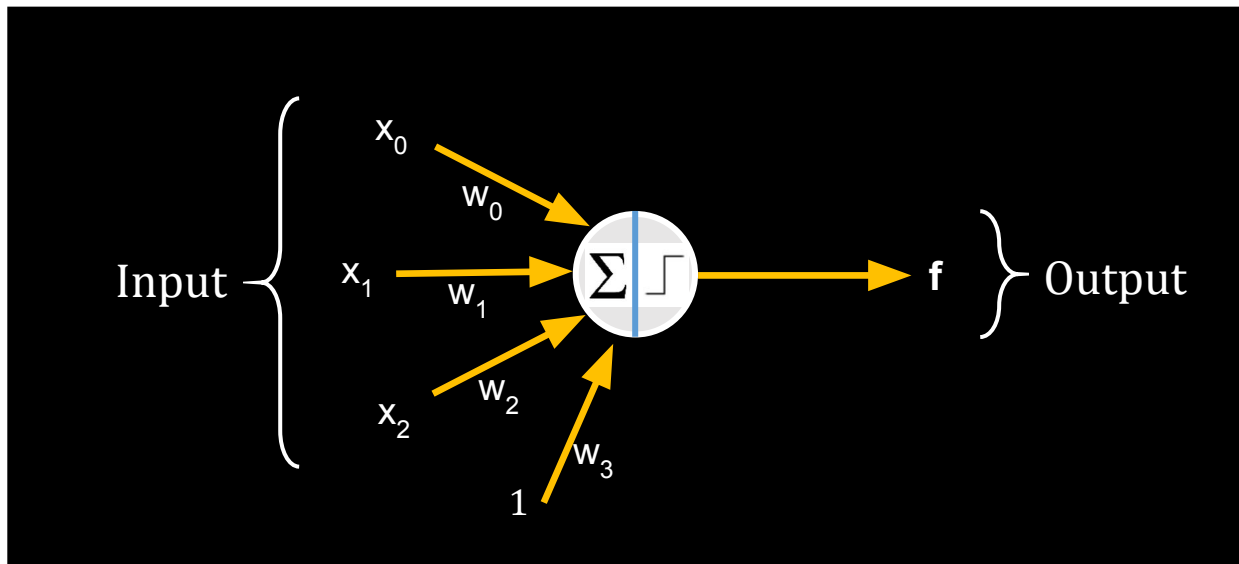


Perceptron

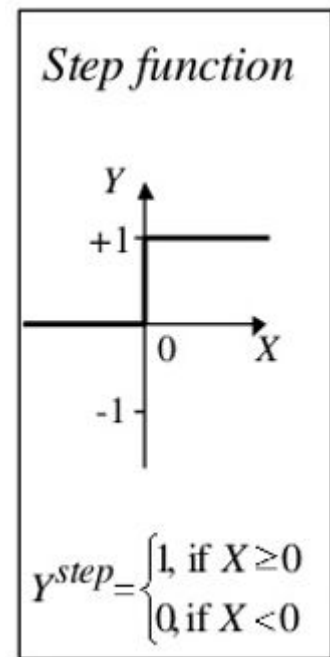


$$x_0w_0 + x_1w_1 + x_2w_2 + w_3$$

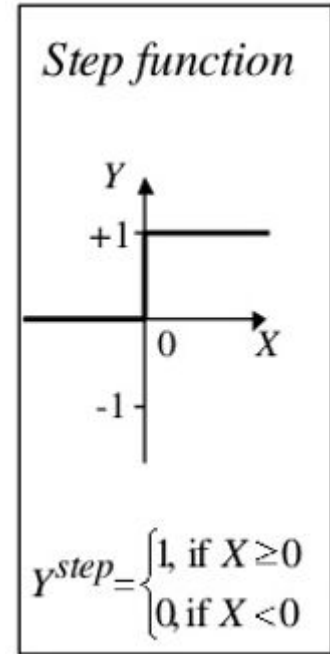
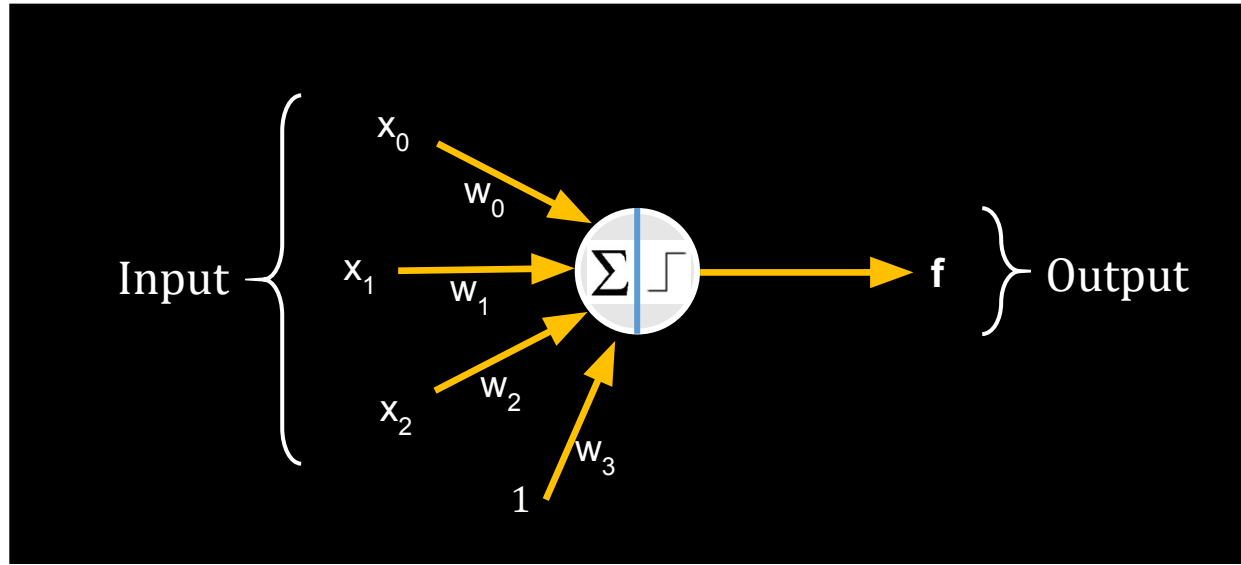
Perceptron



$$x_0w_0 + x_1w_1 + x_2w_2 + w_3$$



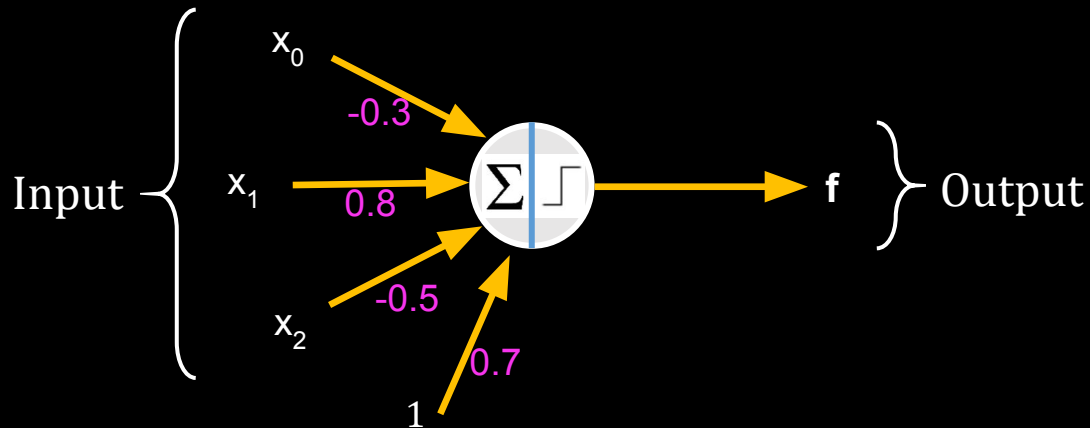
Perceptron



$$f = \begin{cases} 0 & \text{if } x_0w_0 + x_1w_1 + x_2w_2 + w_3 < 0 \\ 1 & \text{if } x_0w_0 + x_1w_1 + x_2w_2 + w_3 \geq 0 \end{cases}$$

Training

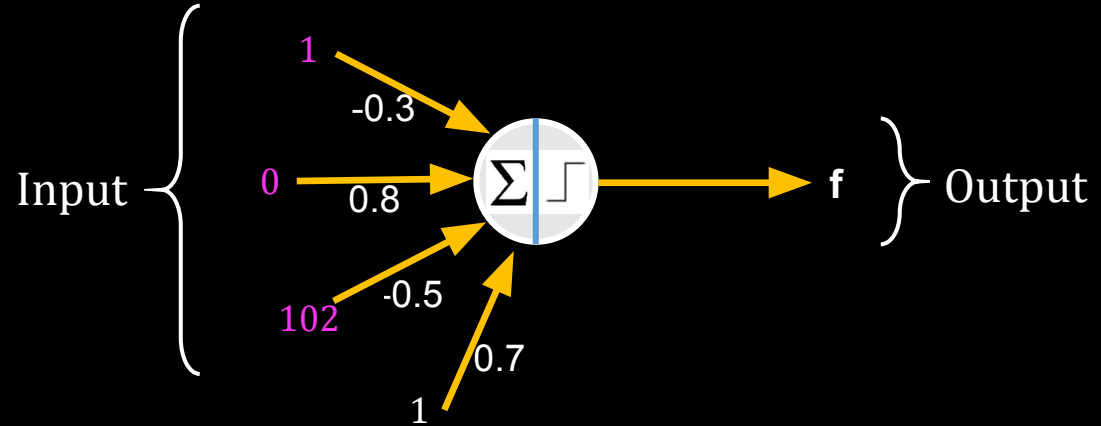
Initialize weights
randomly



Training

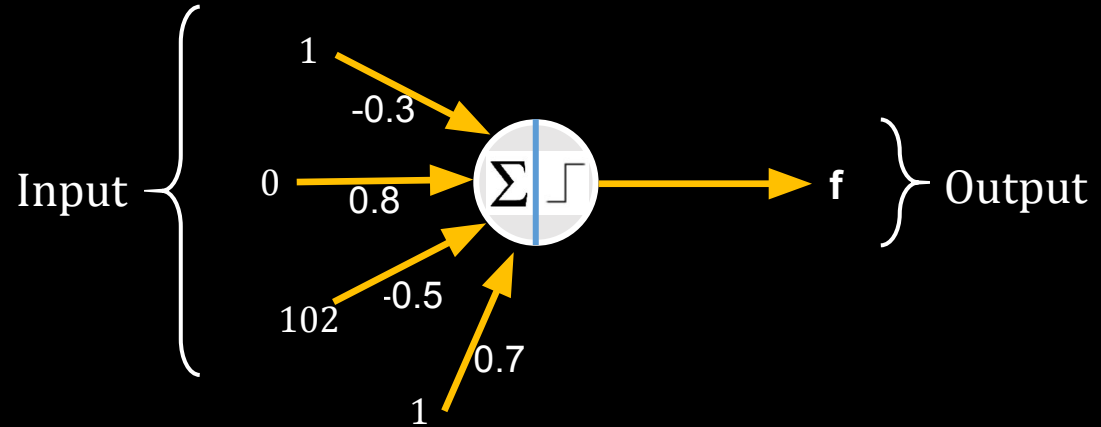
Input a training record

Chills	Congestion	Temp (F)	Flu
Y	N	102	N
Y	Y	98	Y
Y	Y	101	Y
N	Y	99	Y
N	N	97	N
N	Y	98	N
Y	Y	100	Y



Training

Chills	Congestion	Temp (F)	Flu
Y	N	102	N
Y	Y	98	Y
Y	Y	101	Y
N	Y	99	Y
N	N	97	N
N	Y	98	N
Y	Y	100	Y



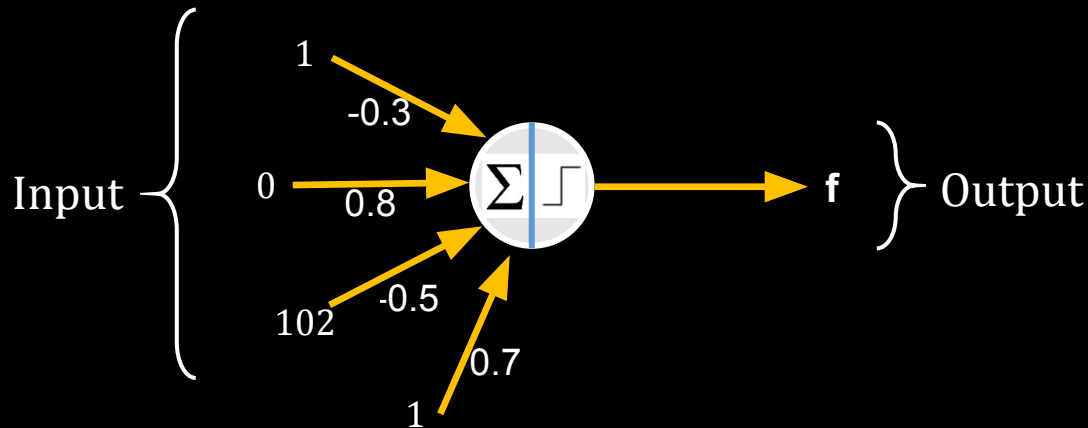
$$-0.3 + 0 - 51 + 0.7 = -50.6$$

$$f = 0$$

Compute the output
(prediction)

Training

Chills	Congestion	Temp (F)	Flu
Y	N	102	N
Y	Y	98	Y
Y	Y	101	Y
N	Y	99	Y
N	N	97	N
N	Y	98	N
Y	Y	100	Y



$$-0.3 + 0 - 51 + 0.7 = -50.6$$

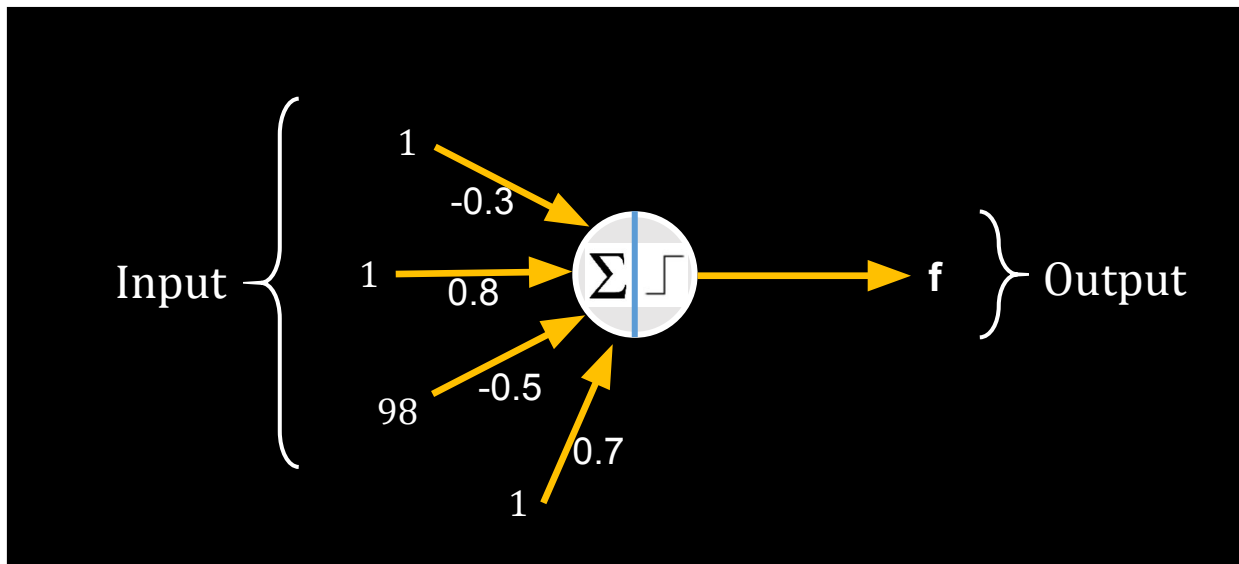
$$f = 0 \checkmark$$

Compare prediction to actual
label

Training

Repeat

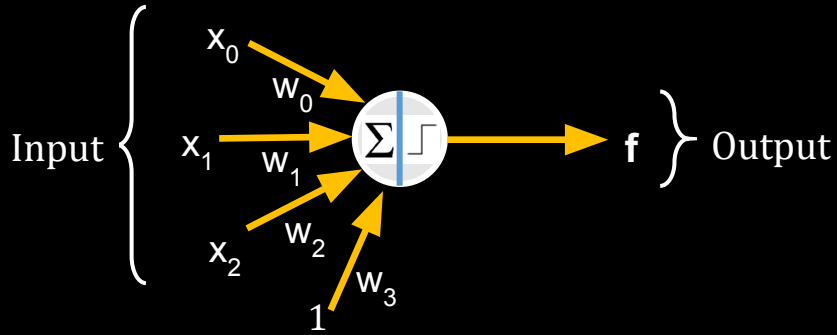
Chills	Congestion	Temp (F)	Flu
Y	N	102	N
Y	Y	98	Y
Y	Y	101	Y
N	Y	99	Y
N	N	97	N
N	Y	98	N
Y	Y	100	Y



$$-0.3 + 0.8 - 49 + 0.7 = -47.8$$

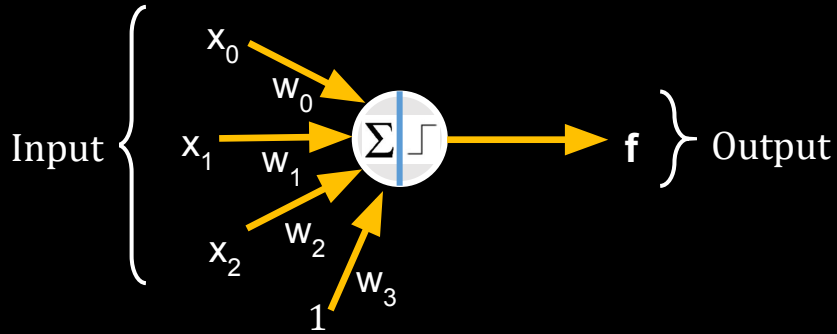
$$f = 0 \quad \text{✗}$$

Updating weights



$$f = \begin{cases} 0 & \text{if } x_0w_0 + x_1w_1 + x_2w_2 + w_3 < 0 \\ 1 & \text{if } x_0w_0 + x_1w_1 + x_2w_2 + w_3 \geq 0 \end{cases}$$

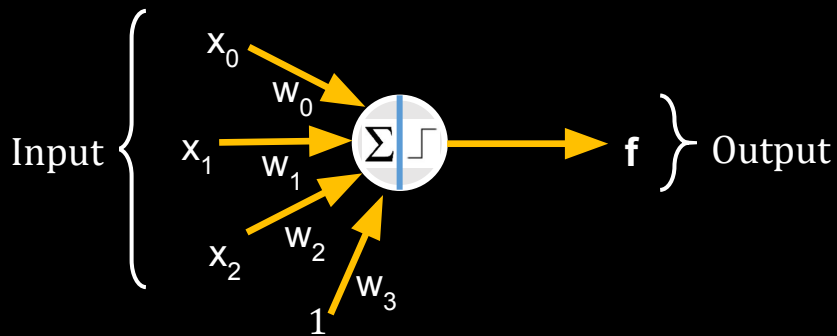
Updating weights



$$w_i = w_i + n(error * x_i)$$

$$f = \begin{cases} 0 & \text{if } x_0w_0 + x_1w_1 + x_2w_2 + w_3 < 0 \\ 1 & \text{if } x_0w_0 + x_1w_1 + x_2w_2 + w_3 \geq 0 \end{cases}$$

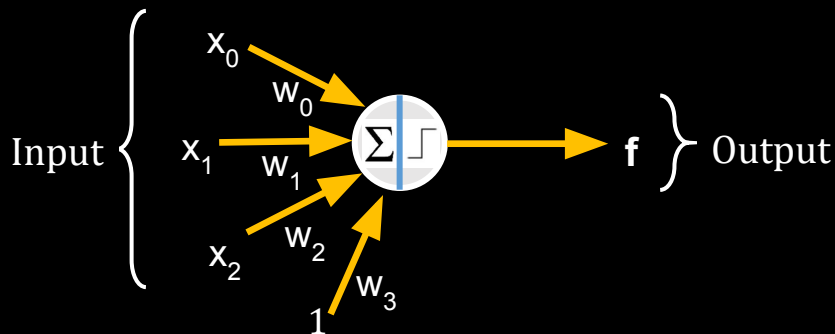
Updating weights



$$w_i = w_i + \cancel{n}^1(error * x_i)$$

$$f = \begin{cases} 0 & \text{if } x_0w_0 + x_1w_1 + x_2w_2 + w_3 < 0 \\ 1 & \text{if } x_0w_0 + x_1w_1 + x_2w_2 + w_3 \geq 0 \end{cases}$$

Updating weights



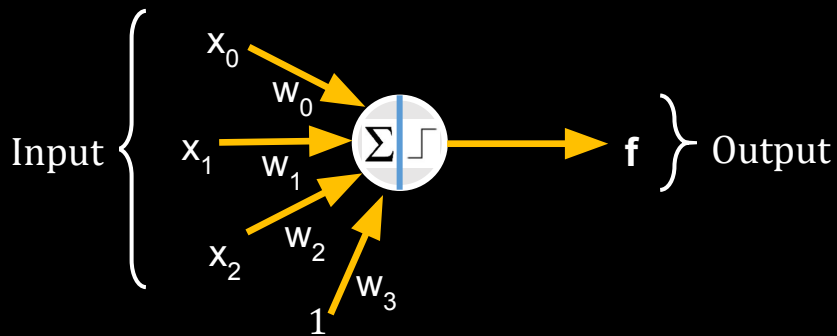
$$f = \begin{cases} 0 & \text{if } x_0w_0 + x_1w_1 + x_2w_2 + w_3 < 0 \\ 1 & \text{if } x_0w_0 + x_1w_1 + x_2w_2 + w_3 \geq 0 \end{cases}$$

$$w_i = w_i + \overset{1}{\cancel{\eta}}(\text{error} * x_i)$$

The error is the difference between the actual label and the predicted label (f).

- If actual=1 and predicted=0, then error = actual-predicted = 1-0 = 1
- If actual=0 and predicted=1, then error = actual-predicted = 0-1 = -1

Updating weights



$$f = \begin{cases} 0 & \text{if } x_0w_0 + x_1w_1 + x_2w_2 + w_3 < 0 \\ 1 & \text{if } x_0w_0 + x_1w_1 + x_2w_2 + w_3 \geq 0 \end{cases}$$

$$w_i = w_i + \overset{1}{\cancel{\eta}}(\text{error} * x_i)$$

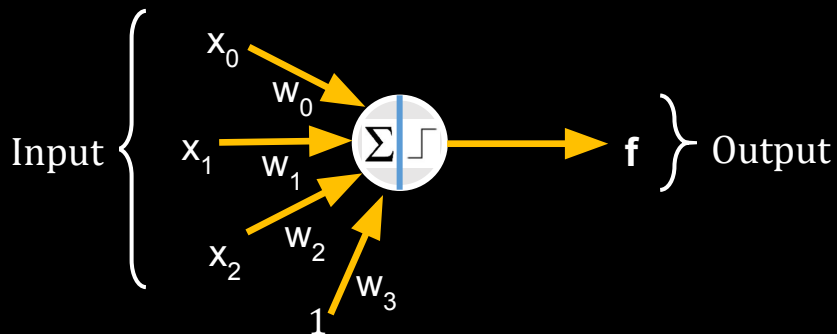
The error is the difference between the actual label and the predicted label (f).

- If actual=1 and predicted=0, then error = actual-predicted = 1-0 = 1
- If actual=0 and predicted=1, then error = actual-predicted = 0-1 = -1

If actual > predicted, then predicted was too low, so we want the weights to go up so that predicted gets higher.

- actual=1, predicted=0
- (actual - predicted) will be positive (+1), so add to the weights to make them go up.

Updating weights



$$f = \begin{cases} 0 & \text{if } x_0w_0 + x_1w_1 + x_2w_2 + w_3 < 0 \\ 1 & \text{if } x_0w_0 + x_1w_1 + x_2w_2 + w_3 \geq 0 \end{cases}$$

$$w_i = w_i + \overset{1}{\cancel{-1}}(\text{error} * x_i)$$

The error is the difference between the actual label and the predicted label (f).

- If actual=1 and predicted=0, then error = actual-predicted = 1-0 = 1
- If actual=0 and predicted=1, then error = actual-predicted = 0-1 = -1

If actual > predicted, then predicted was too low, so we want the weights to go up so that predicted gets higher.

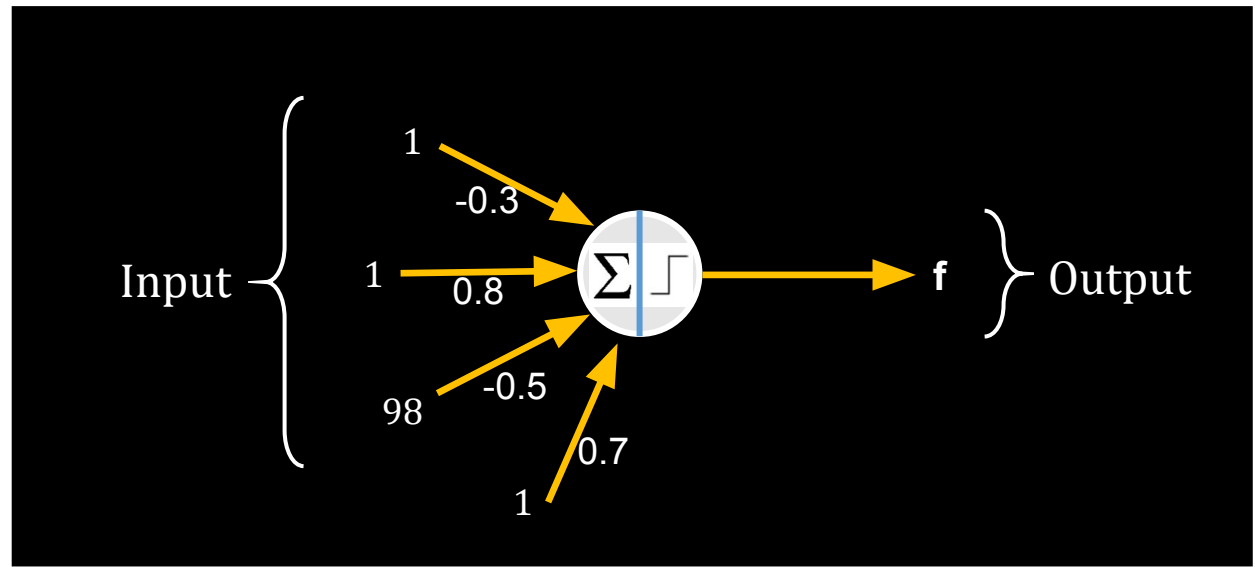
- actual=1, predicted=0
- (actual - predicted) will be positive (+1), so add to the weights to make them go up.

If actual < predicted, then predicted was too high, so we want the weights to go down so that predicted gets lower.

- actual=0, predicted=1
- (actual - predicted) will be negative (-1), so subtract from the weights to make them go down

Training

Chills	Congestion	Temp (F)	Flu
Y	N	102	N
Y	Y	98	Y
Y	Y	101	Y
N	Y	99	Y
N	N	97	N
N	Y	98	N
Y	Y	100	Y

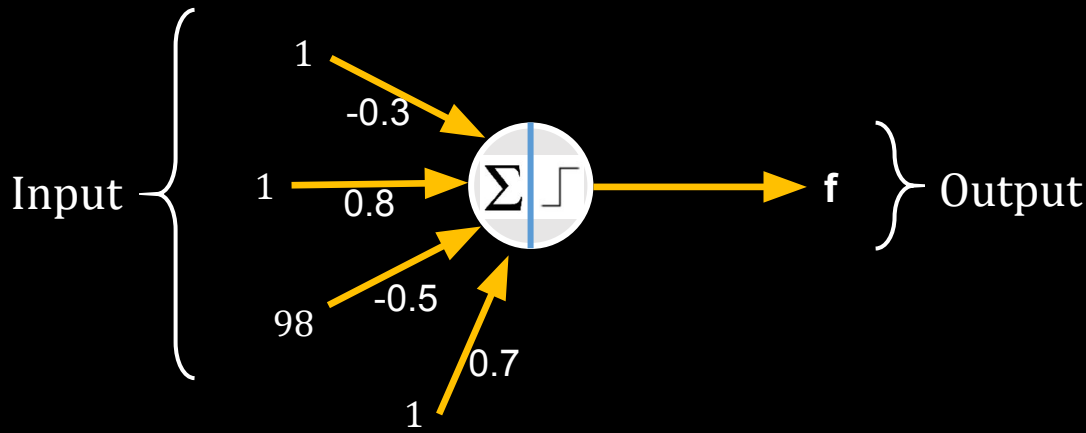


$$-0.3 + 0.8 - 49 + 0.7 = -47.8$$

$$f = 0 \quad \text{X}$$

Training

Chills	Congestion	Temp (F)	Flu
Y	N	102	N
Y	Y	98	Y
Y	Y	101	Y
N	Y	99	Y
N	N	97	N
N	Y	98	N
Y	Y	100	Y

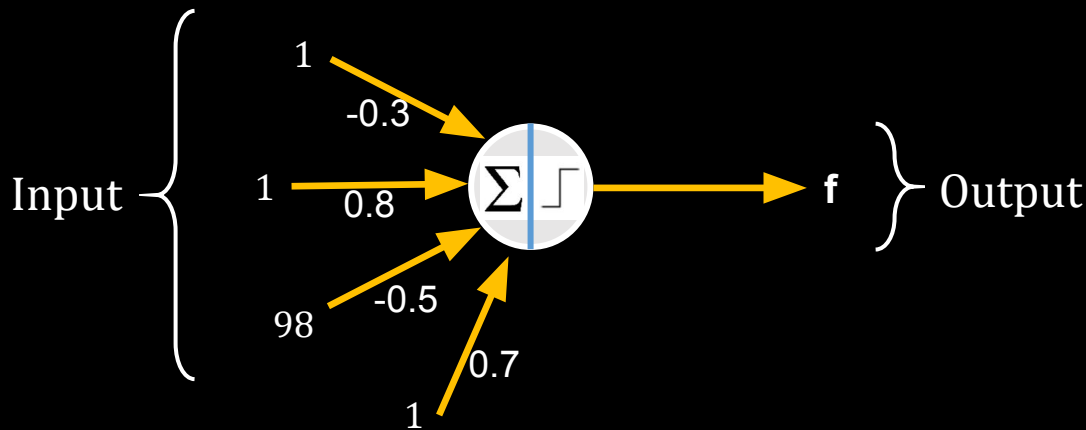


$$-0.3 + 0.8 - 49 + 0.7 = -47.8$$

$$f = 0 \quad \text{Error} = 1 - 0 = 1$$

Training

Chills	Congestion	Temp (F)	Flu
Y	N	102	N
Y	Y	98	Y
Y	Y	101	Y
N	Y	99	Y
N	N	97	N
N	Y	98	N
Y	Y	100	Y



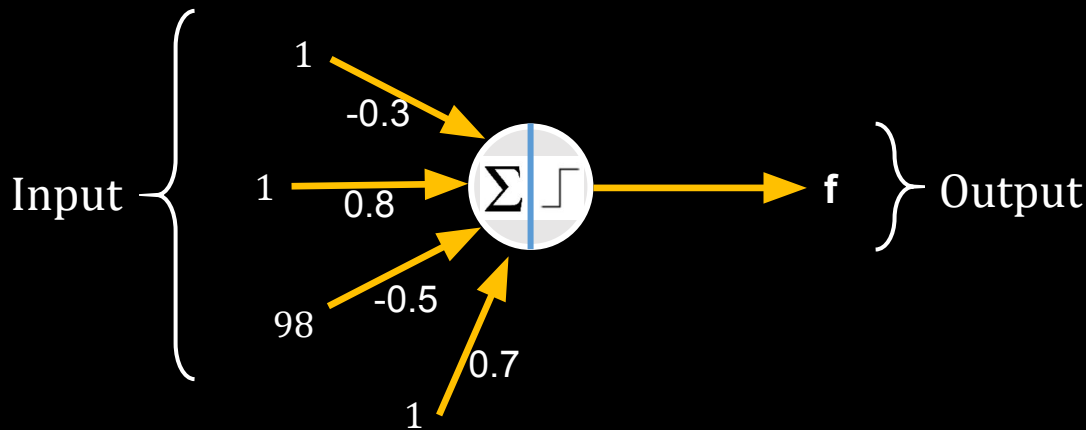
$$-0.3 + 0.8 - 49 + 0.7 = -47.8$$

$$f = 0 \quad \text{Error} = 1 - 0 = 1$$

$$w_0 = -0.3 + (1 * 1) = 0.7$$

Training

Chills	Congestion	Temp (F)	Flu
Y	N	102	N
Y	Y	98	Y
Y	Y	101	Y
N	Y	99	Y
N	N	97	N
N	Y	98	N
Y	Y	100	Y



$$-0.3 + 0.8 - 49 + 0.7 = -47.8$$

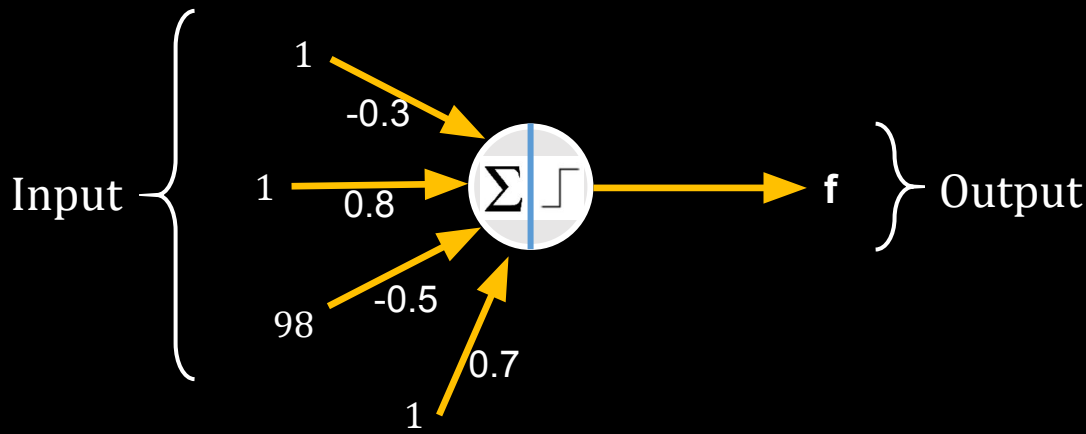
$$f = 0 \quad \text{Error} = 1 - 0 = 1$$

$$w_0 = -0.3 + (1 * 1) = 0.7$$

$$w_1 = 0.8 + (1 * 1) = 1.8$$

Training

Chills	Congestion	Temp (F)	Flu
Y	N	102	N
Y	Y	98	Y
Y	Y	101	Y
N	Y	99	Y
N	N	97	N
N	Y	98	N
Y	Y	100	Y



$$-0.3 + 0.8 - 49 + 0.7 = -47.8$$

$$f = 0 \quad \text{Error} = 1 - 0 = 1$$

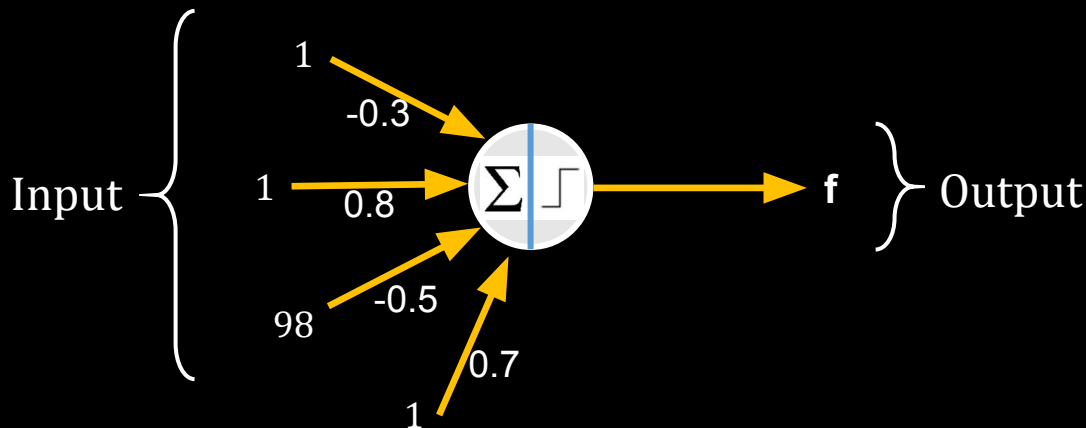
$$w_0 = -0.3 + (1 * 1) = 0.7$$

$$w_1 = 0.8 + (1 * 1) = 1.8$$

$$w_2 = -0.5 + (1 * 98) = 97.5$$

Training

Chills	Congestion	Temp (F)	Flu
Y	N	102	N
Y	Y	98	Y
Y	Y	101	Y
N	Y	99	Y
N	N	97	N
N	Y	98	N
Y	Y	100	Y



$$-0.3 + 0.8 - 49 + 0.7 = -47.8$$

$$f = 0 \quad \text{Error} = 1 - 0 = 1$$

$$w_0 = -0.3 + (1 * 1) = 0.7$$

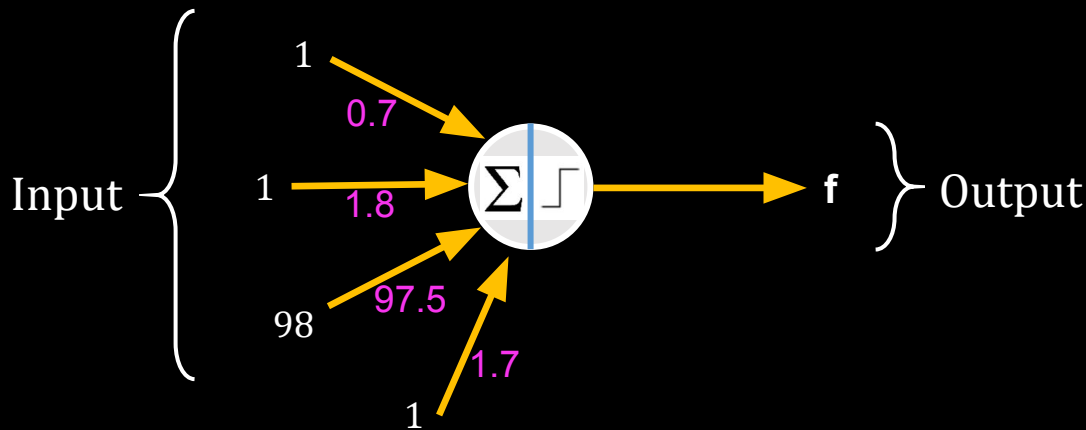
$$w_1 = 0.8 + (1 * 1) = 1.8$$

$$w_2 = -0.5 + (1 * 98) = 97.5$$

$$w_3 = 0.7 + (1 * 1) = 1.7$$

Training

Chills	Congestion	Temp (F)	Flu
Y	N	102	N
Y	Y	98	Y
Y	Y	101	Y
N	Y	99	Y
N	N	97	N
N	Y	98	N
Y	Y	100	Y



$$-0.3 + 0.8 - 49 + 0.7 = -47.8$$

$$f = 0 \quad \text{Error} = 1 - 0 = 1$$

$$w_0 = -0.3 + (1 * 1) = 0.7$$

$$w_1 = 0.8 + (1 * 1) = 1.8$$

$$w_2 = -0.5 + (1 * 98) = 97.5$$

$$w_3 = 0.7 + (1 * 1) = 1.7$$

